harf

num

$\underline{25} \times \underline{25} \mid \times \underline{10} \times \underline{9} \times \underline{8} =$

①

$\underline{8}\ \underline{7}\ \underline{6}\ \underline{5}\ \underline{4}\ \underline{3}\ \underline{2}\ \underline{1} = 8!$

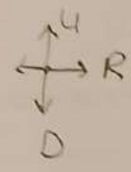
$8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

②

A(2,1)

B(7,4)

$5!.3! \;/\; \frac{8!}{5!.3!} \;/$



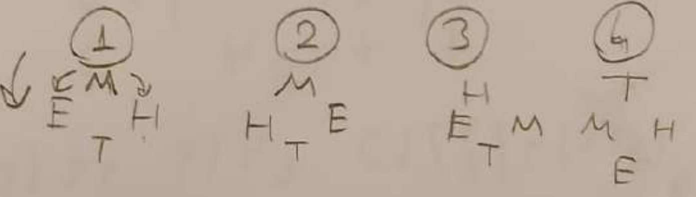
$\rightarrow (\bar{5},3) \Rightarrow 8$

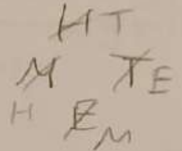
$\Rightarrow \frac{8!}{5!.3!} = \frac{87\cancel{6}}{\cancel{6}} = 56$

③

$(4-2)! = ⑥$

Eren, Muhammed, Talha, Hasan





$6! = 24$

$\frac{24}{4=n} = 6$

④

⑤

Number of Combinations Examples

- for a volleyball team, gym teacher must Select nine girls from junior and Senior classes
- 28 junior and 25 Senior candidates
- How many different ways?
- if no restrictions: $\left(\frac{53}{9}\right) = 4,437,613,550$
- if two juniors and one Senior are best spikers and must be on the team $\binom{50}{6} = 15,890,700$
- if there has to be for juniors and five Seniors: $\binom{28}{4}\binom{25}{5} = 1,087,836,750$

$\boxed{C(28,4)^{\circ}\ C(25,5)}$

$$(a+b)^4 = \binom{4}{0} a^4 b^0 + \overset{4}{\binom{4}{1}} a^3 b$$

$$\rightarrow \underbrace{\binom{4}{2}}_{6} a^2 b^2 + \underbrace{\binom{4}{3}}_{4} ab^3 + \binom{4}{4} a^0 b^4$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1

⑥

a) 7^4 b) 4^7 c) $\binom{7}{4} 4!$ d) e)

$$\frac{n!}{n} = \frac{(n-1)! \cdot \cancel{n}}{\cancel{n}}$$

c h t f
xx | xx | xx | x
★★★★ | | ★★★ |

xx -- x 7 çiz, → 1 ürün

4 ürün
$\overbrace{\text{4çiz, 3tako}}^{\text{2ürün}}$, 2çiz, 2balık, 2tako
/ | \ \
A⊖A⊖A⊖A

1 hotdog

⑦

Combinations with Repetition Example

$n=4, r=7$ r arrangement'ı yapılacak nesne

cchhttf	xx|xx|xx|x
cccchtf	xxxx|x|x|x
ccccccf	xxxxxx|||x
httffff	|x|xx|xxxx
ttttttt	||xxxxxxx|
fffffff	|||xxxxxxx

⑧